

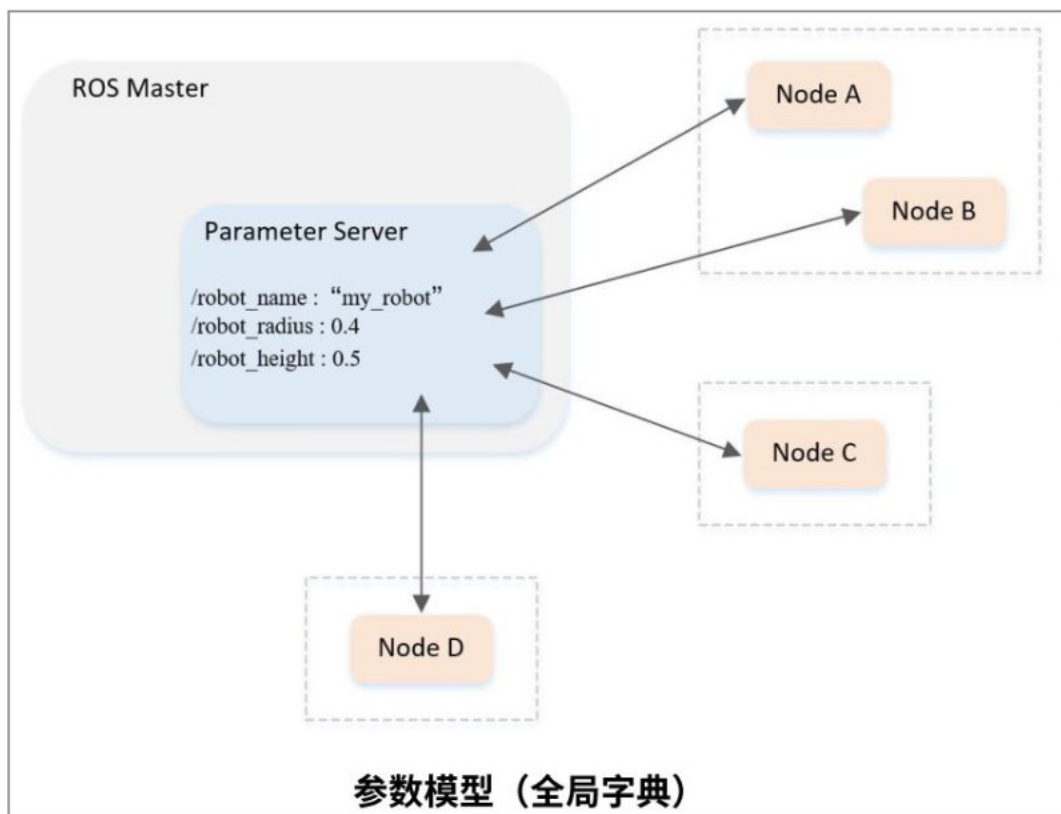
第 11 课 参数的使用与编程方法

1.服务模型

在ROS master当中有一个参数服务器，它是一个全局字典，用来保存各个节点之间的配置参数的。例如里面有Parameter server保存我们的名字、半径和高度，是各个节点都可以全局访问的。

比如说我在Node A点里面来访问robot name，然后就会获取得到一个“my_rot”的值，它只要给我们的ROS master去发送一个查询的请求，然后就会返回“my_rot”的结果。Node B、Node C和Node D也是一样。

Parameter server参数服务器模型如下图所示：



2. rosparam参数

2.1 rosparam详细参数

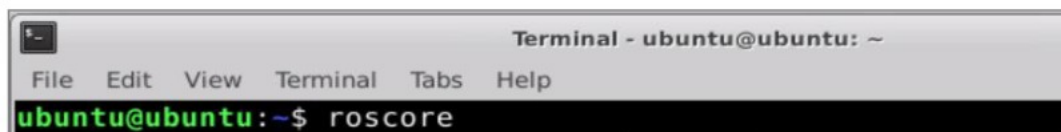
这里我们先来了解下rosparam参数，详细参数如下图：

- 列出当前多有参数
`$ rosparam list`
- 显示某个参数值
`$ rosparam get param_key`
- 设置某个参数值
`$ rosparam set param_key param_value`
- 保存参数到文件
`$ rosparam dump file_name`
- 从文件读取参数
`$ rosparam load file_name`
- 删除参数
`$ rosparam delete param_key`

2.2 运行海龟例程

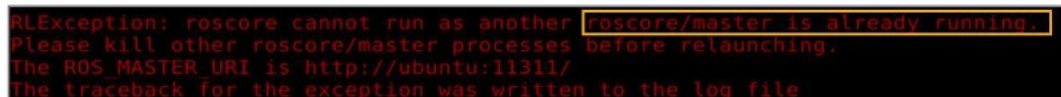
这里以海龟案例为例，先运行海龟例程，步骤如下：

- 1) 输入指令“roscore”，回车。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roscore
```

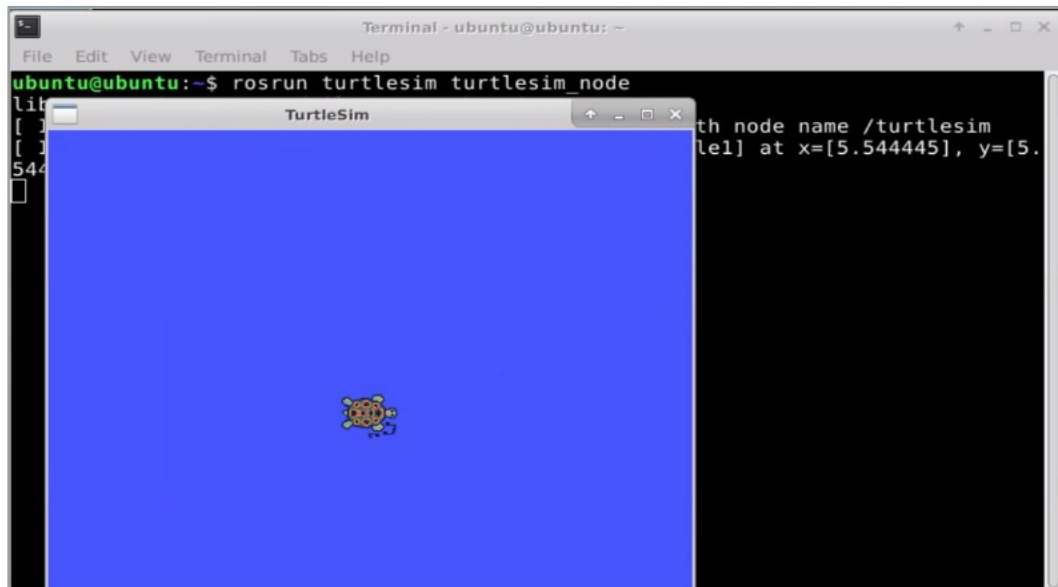
若已开启，则会出现以下提示：



```
RLException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://ubuntu:11311/
The traceback for the exception was written to the log file
```



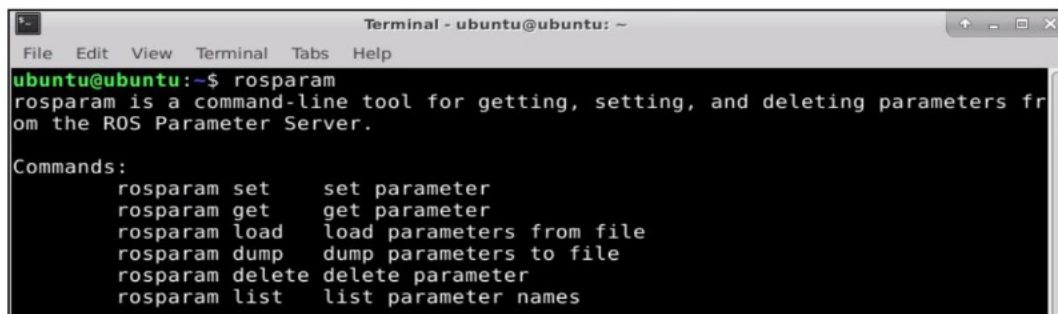
- 2) 输入指令“**roslaunch turtlesim turtlesim_node**”，回车，打开小海龟的仿真器。



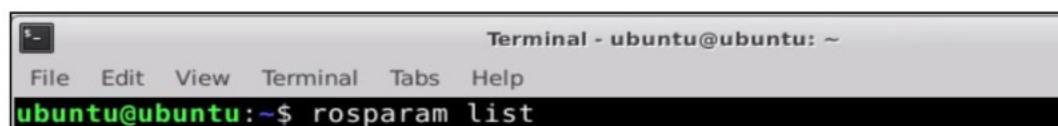
2.3 rosparam的使用

再打开一个终端， rosparam的使用，步骤如下：

- 1) 输入指令“**rosparam**”，回车。



- 2) 输入指令“**rosparam list**”，回车，查询海龟参数个数。



3) 输入指令“`rosparam get /turtlesim/background_b`”，回车，获取“`background_b`”数值，其他数值获取方法类似。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosparam get /turtlesim/background_b
255
```

4) 输入指令“`rosparam set /turtlesim/background_b 100`”，回车，设置“`background_b`”数值，其他数值设置方法类似。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosparam set /turtlesim/background_b 100
```

5) 输入指令“`rosparam get /turtlesim/background_b`”，回车，看到数值改为100了。

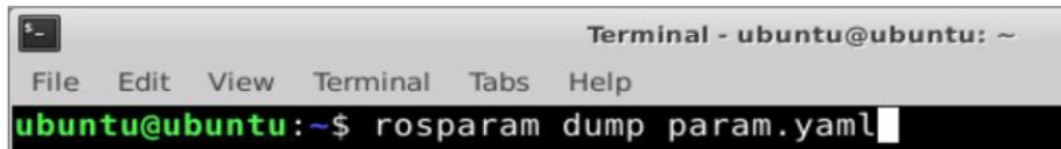
```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosparam set /turtlesim/background_b 100
ubuntu@ubuntu:~$ rosparam get /turtlesim/background_b
100
```

6) 输入指令“`rosservice call clear "{}"`”，回车，发送请求，更改颜色。

```
ubuntu@ubuntu:~$ rosservice call clear "{}"
```

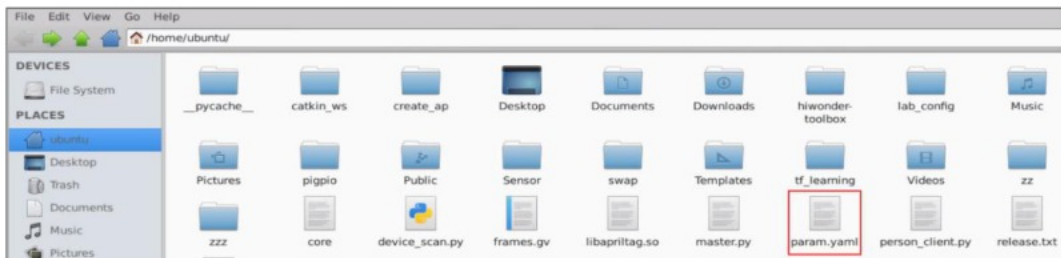


- 7) 输入指令“`rosparam dump param.yaml`”，回车，保存到文件。

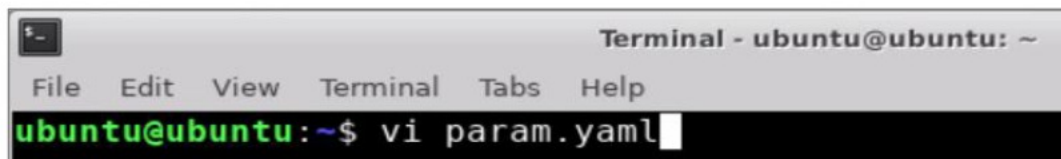


```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosparam dump param.yaml
```

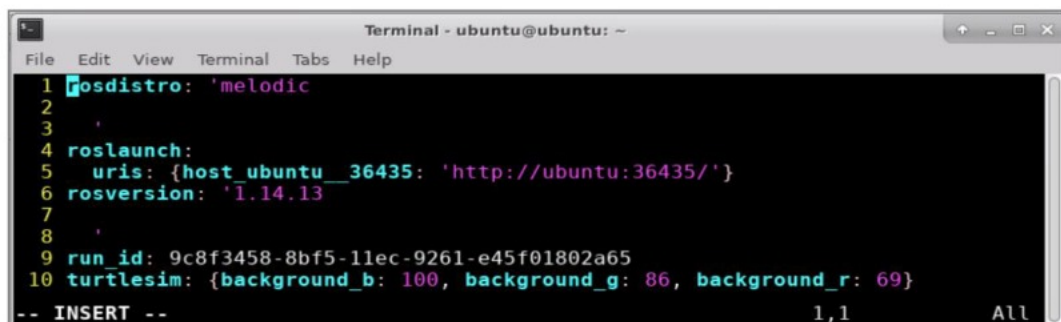
- 8) 文件打开在默认路径，我们保存在主文件路径，直接打开。



- 1) 输入指令“`vi param.yaml`”，回车。按下“i”修改“`param.yaml`”。

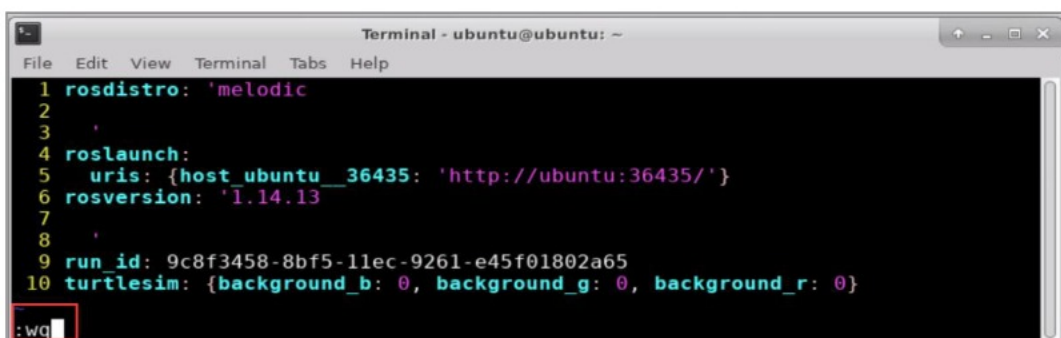


```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ vi param.yaml
```



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
1 rosdistro: 'melodic'
2
3
4 roslaunch:
5   uris: {host_ubuntu__36435: 'http://ubuntu:36435/'}
6 rosversion: '1.14.13'
7
8
9 run_id: 9c8f3458-8bf5-11ec-9261-e45f01802a65
10 turtlesim: {background_b: 100, background_g: 86, background_r: 69}
-- INSERT --                               1,1      All
```

- 9) 可以将颜色都改为0，也就是黑色，修改完成，按下“Esc”，输入“`:wq`”保存并退出。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
1 rosdistro: 'melodic'
2
3
4 roslaunch:
5   uris: {host_ubuntu__36435: 'http://ubuntu:36435/'}
6 rosversion: '1.14.13'
7
8
9 run_id: 9c8f3458-8bf5-11ec-9261-e45f01802a65
10 turtlesim: {background_b: 0, background_g: 0, background_r: 0}
:wq
```



10) 输入指令“**rosparam load param.yaml**”，回车，加载文件。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosparam load param.yaml
```

11) 输入指令“**rosparam get /turtlesim/background_b**”，回车，检查加载效果。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosparam load param.yaml
ubuntu@ubuntu:~$ rosparam get /turtlesim/background_b
0
```

12) 输入指令“**rosservice call clear "{}"**”，回车，发送请求，更改颜色，背景颜色为黑色。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosservice call clear "{}"
```



13) 输入指令“**rosparam delete /turtlesim/background_g**”，回车，删除g的颜色。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosparam delete /turtlesim/background_g
```



14) 输入指令“**rosparam list**”，回车，查看效果。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosparam list
```

15) 输入指令“**rosservice call clear "{}"**”，回车，重新刷新背景颜色，查看效果。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosservice call clear "{}"
```



3.编程方法

创建功能包，步骤如下：

1) 输入指令“**cd catkin_ws/src/**”，回车，进入工作空间。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ cd catkin_ws/src/
```

输入指令“**catkin_create_pkg parameter_hiwonder rospy std_msgs**”，回车，创建功能包。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ cd catkin_ws/src/
ubuntu@ubuntu:~/catkin_ws/src$ catkin_create_pkg parameter_hiwonder rospy std_ms
gs
```

3.1 编写控制程序

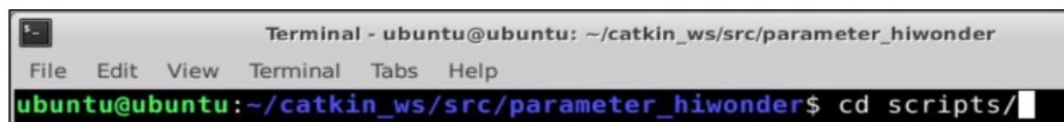
- 1) 打开Linux命令行终端。
- 2) 输入进入功能包所在目录的指令“`roscd parameter_hiwonder`”，并按下回车。

```
hiwonder@ubuntu:~$ roscd parameter_hiwonder
```

- 3) 输入指令“`mkdir scripts`”，并按下回车，新建用于存放Python脚本的文件夹“`scripts`”。

```
hiwonder@ubuntu:~/catkin_ws/src/parameter_hiwonder$ mkdir scripts
```

- 4) 输入指令“`cd scripts/`”，并按下回车，进入用于存放Python脚本的文件夹“`scripts`”。



A terminal window titled "Terminal - ubuntu@ubuntu: ~/catkin_ws/src/parameter_hiwonder" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The command prompt shows "ubuntu@ubuntu:~/catkin_ws/src/parameter_hiwonder\$ cd scripts/" with the cursor at the end of the line.

- 5) 输入指令“`vi parameter_config.py`”编辑程序，复制下面程序。如需修改，再按下“`i`”即可修改。修改完成，按下“`Esc`”，输入“`: wq`”保存并退出。



A terminal window titled "Terminal - ubuntu@ubuntu: ~/catkin_ws/src/parameter_hiwonder/scripts" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The command prompt shows "ubuntu@ubuntu:~/catkin_ws/src/parameter_hiwonder/scripts\$ vi parameter_config.py" with the cursor at the end of the line.

```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

# 该例程设置/读取海龟例程中的参数

import sys

import rospy

from std_srvs.srv import Empty
```

```
def parameter_config():

    # ROS节点初始化

    rospy.init_node('parameter_config', anonymous=True)

    # 读取背景颜色参数

    red    = rospy.get_param('/turtlesim/background_r')

    green = rospy.get_param('/turtlesim/background_g')

    blue   = rospy.get_param('/turtlesim/background_b')

    rospy.loginfo("Get Backgroud Color[%d, %d, %d]", red, green, blue)

    # 设置背景颜色参数

    rospy.set_param("/turtlesim/background_r", 255);

    rospy.set_param("/turtlesim/background_g", 255);

    rospy.set_param("/turtlesim/background_b", 255);

    rospy.loginfo("Set Backgroud Color[255, 255, 255]");

    # 读取背景颜色参数

    red    = rospy.get_param('/turtlesim/background_r')

    green = rospy.get_param('/turtlesim/background_g')

    blue   = rospy.get_param('/turtlesim/background_b')
```

```
rospy.logininfo("Get Backgroud Color[%d, %d, %d]", red, green, blue)

# 发现/spawn服务后，创建一个服务客户端，连接名为/spawn的service

rospy.wait_for_service('/clear')

try:

    clear_background = rospy.ServiceProxy('/clear', Empty)

    # 请求服务调用，输入请求数据

    response = clear_background()

    return response

except rospy.ServiceException:

    pass

if __name__ == "__main__":

    parameter_config()
```



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/parameter_hiwonder/scripts
File Edit View Terminal Tabs Help
25
26 rospy.loginfo("Set Background Color[255, 255, 255]");
27
28 # 读取背景颜色参数
29 red = rospy.get_param('/turtlesim/background_r')
30 green = rospy.get_param('/turtlesim/background_g')
31 blue = rospy.get_param('/turtlesim/background_b')
32
33 rospy.loginfo("Get Background Color[%d, %d, %d]", red, green, blue)
34
35 # 发现/spawn服务后，创建一个服务客户端，连接名为/spawn的 service
36 rospy.wait_for_service('/clear')
37 try:
38     clear_background = rospy.ServiceProxy('/clear', Empty)
39
40     # 请求服务调用，输入请求数据
41     response = clear_background()
42     return response
43 except rospy.ServiceException, e:
44     print "Service call failed: %s"%e
45
46 if __name__ == "__main__":
47     parameter_config()
:wq
```

6) 输入指令“**chmod +x parameter_config.py**”回车，为保存的parameter_config.py赋予可执行权限。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/parameter_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/parameter_hiwonder/scripts$ chmod +x parameter_config.py
```

4.运行程序

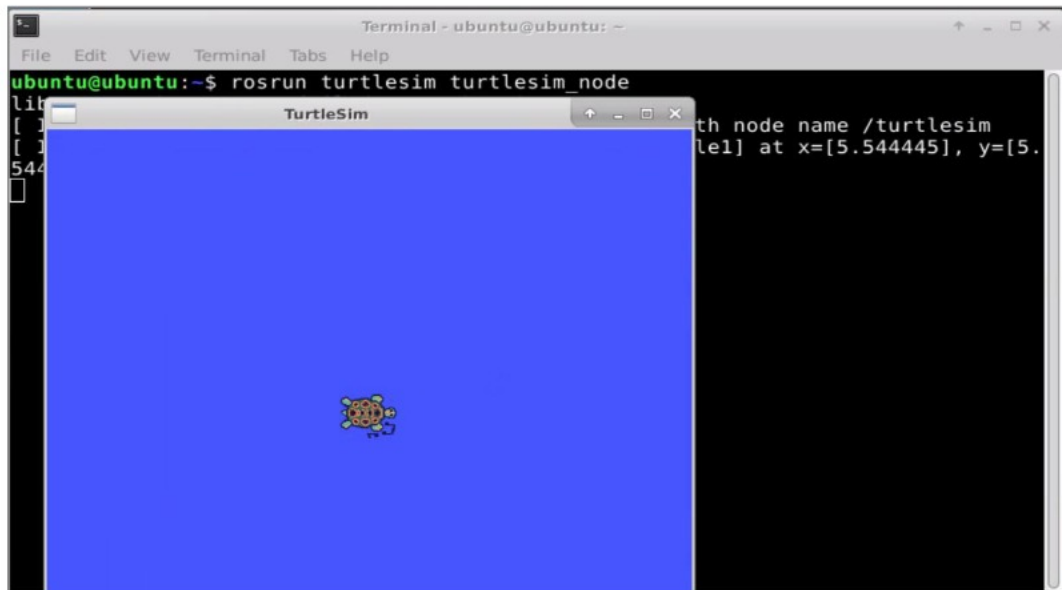
1) 输入指令“**roscore**”，并按下回车，启动节点管理器。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ roscore
```

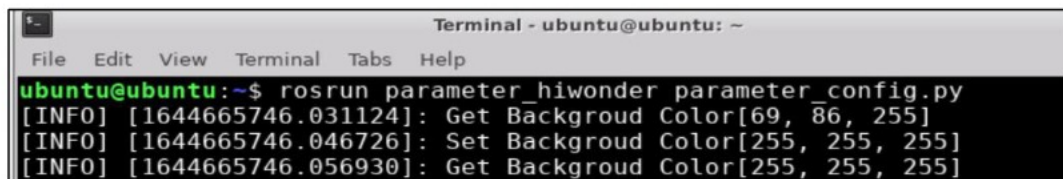
若已开启，则会出现以下提示：

```
RLException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://ubuntu:11311/
The traceback for the exception was written to the log file
```

- 2) 输入指令“**roslaunch turtlesim turtlesim_node**”，回车，打开小海龟的仿真器。



- 3) 输入指令“**roslaunch parameter_hiwonder parameter_config.py**”，并按下回车，运行程序，它将海龟的背景颜色改为白色，显示如下图：



- 4) 如需停止程序，可按下快捷键“**Ctrl+C**”。